

Session 2- Part 1: UX Design Principles

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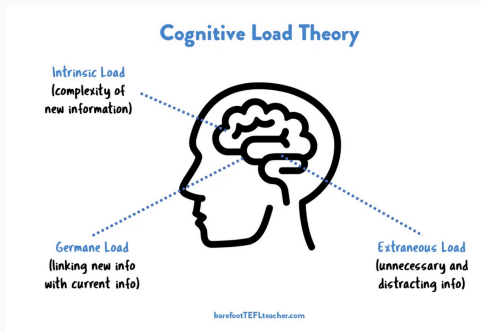
Otago Polytechnic, Auckland International Campus (AIC)

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Cognitive Load

- Users often perceive aesthetically pleasing designs as more usable.
- A well-designed interface that is visually appealing can make users feel that it is easier to use, even if it's not.
- A well-designed app with a clean interface often gets positive user feedback and higher ratings compared to a cluttered one, even if both have the same functionality.
- Focus on visual appeal to enhance perceived usability.

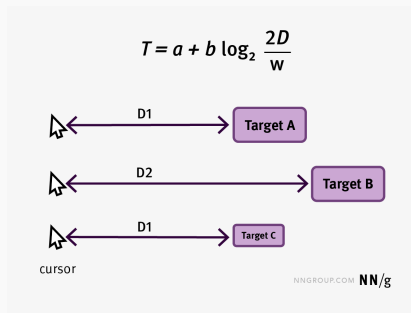


Doherty Threshold

- Productivity increases when the interaction speed between user and computer is less than 400ms.
- If the system responds quickly, users can work more efficiently.
- A website that loads search results almost instantly versus one with noticeable delays.
- Optimize system performance to keep response times under 400ms.

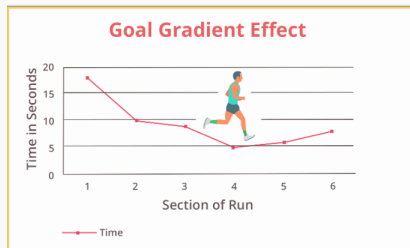
Fitts's Law

- 1 The time to acquire a target is a function of the distance to and size of the target.
- 2 Larger and closer targets are easier and quicker to interact with.
- 3 Large buttons on a mobile app versus small, hard-to-tap buttons.
- 4 Design interactive elements to be large and easily accessible.



Goal-Gradient Effect's Law

- 1 The tendency to work harder as one gets closer to a goal.
- 2 Users are more motivated to complete tasks as they approach their goal.
- 3 Example: A progress bar that shows how much is left to complete a task versus one that doesn't.
- 4 Design progress indicators to encourage user engagement as they approach their goals.



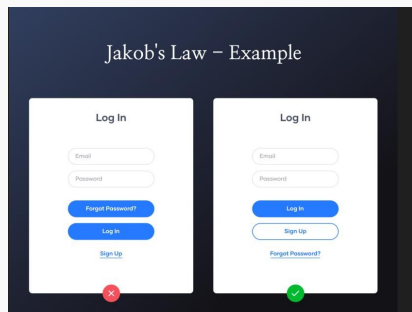
Hick's Law

- 1 The time to make a decision increases with the number and complexity of choices.
- 2 More choices lead to longer decision-making times.
- 3 Example: A menu with many options versus a streamlined menu with fewer, more relevant options.
- 4 Reduce the number of choices to simplify decision-making.



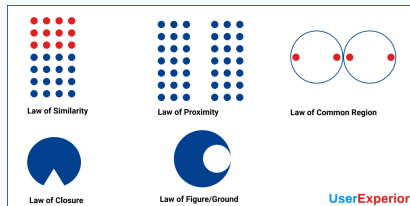
Jakob's Law

- 1 Users spend most of their time on other sites and prefer your site to work similarly.
- 2 Consistency with other sites can make your interface more intuitive.
- 3 Example: Common design patterns such as standard navigation bars and search functions.
- 4 Follow common design patterns to align with user expectations.



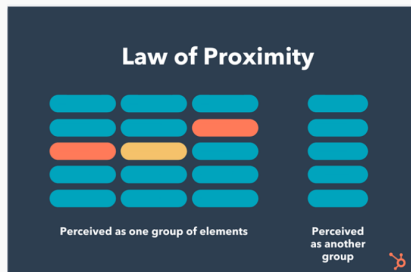
Law of Common Region

- 1 Elements are perceived as grouped if they share a common boundary.
- 2 Defining boundaries helps users recognize related elements.
- 3 Example: A form with grouped sections inside boxes versus a form without visual grouping.
- 4 Use visual grouping to indicate related items.



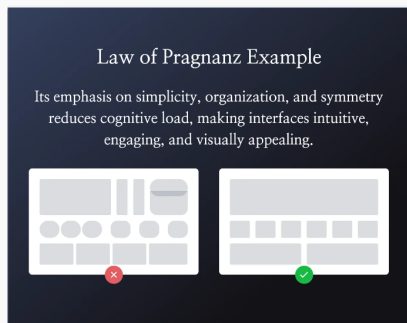
Law of Proximity

- 1 Objects near each other are perceived as grouped together.
- 2 Proximity affects how users group and interpret elements.
- 3 Example: A list of features with related items close together versus items scattered.
- 4 Arrange related items close together to aid in recognition.



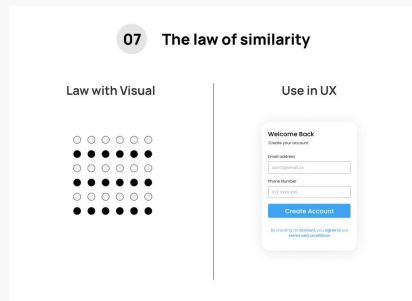
Law of Prägnanz

- 1 People interpret ambiguous or complex images in the simplest form.
- 2 Designs should aim for simplicity to be easily understood.
- 3 Example: A simplified logo design versus a complex and detailed one.
- 4 Strive for clarity and simplicity in design.



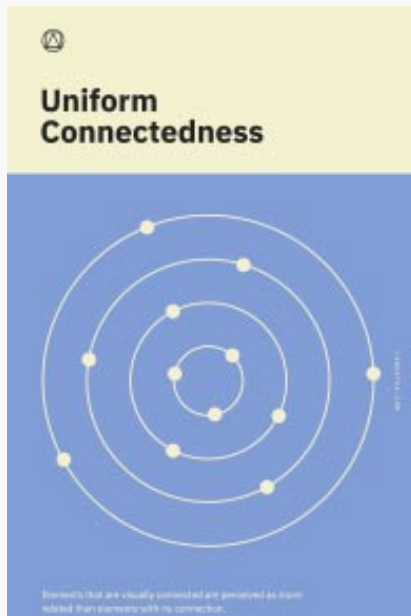
Law of Similarity

- 1 Similar elements are perceived as a group, even if separated.
- 2 Use similarity in color, shape, or size to create visual grouping.
- 3 Example: Similar buttons with the same color for related actions versus buttons with different colors.
- 4 Leverage similarity to create cohesive and intuitive interfaces.



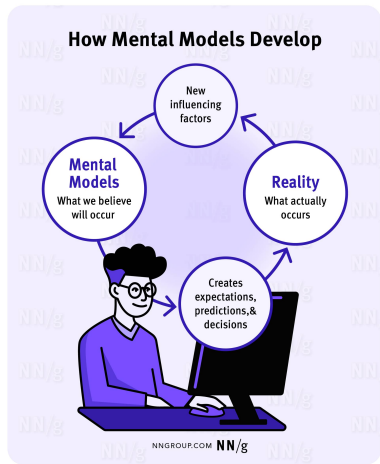
Law of Uniform Connectedness

- 1 Elements that are visually connected are perceived as related.
- 2 Visual connections indicate relationships between elements.
- 3 Example: A dashboard with connected sections versus disconnected sections.
- 4 Use visual cues like lines or backgrounds to connect related elements.



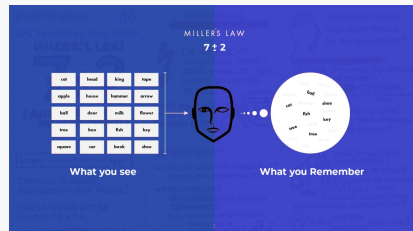
Mental Model

- 1 A compressed model based on what we think we know about a system.
- 2 Users form mental models of how systems work based on their experiences.
- 3 Example: Expecting a trash can icon to delete items based on common conventions.
- 4 Design interfaces that align with users' existing mental models.



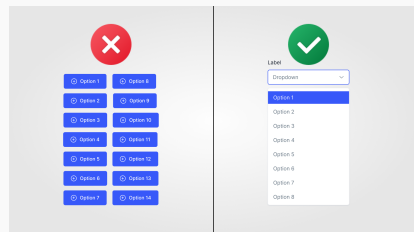
Miller's Law

- 1 The average person can keep $7 (\pm 2)$ items in their working memory.
- 2 Limit the number of items presented to avoid overwhelming users.
- 3 Example: A navigation menu with 5 options versus 10 options.
- 4 Keep information and choices within the limits of working memory.



Occam's Razor

- 1 The simplest solution is usually the best.
- 2 Favor solutions with the fewest assumptions.
- 3 Example: A straightforward user interface versus one with unnecessary complexity.
- 4 Opt for simpler solutions that achieve the desired outcome.

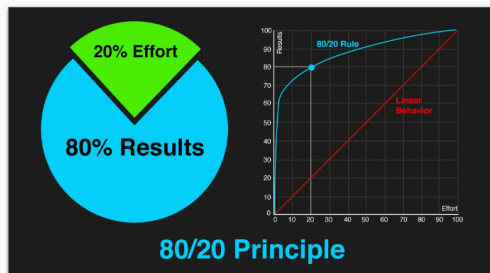


Paradox of the Active User

- 1 Users start using software immediately and rarely read manuals.
- 2 Users prefer to learn by exploring rather than reading instructions.
- 3 Example: Software with intuitive self-explanatory features versus those requiring extensive manuals.
- 4 Design intuitive interfaces that facilitate learning through use.

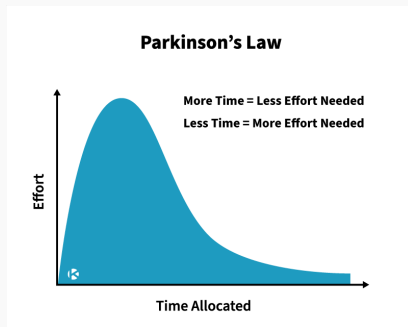
Pareto Principle

- 1 80% of effects come from 20% of causes.
- 2 Focus on the most impactful elements for maximum results.
- 3 Example: 20% of features driving 80% of user engagement.
- 4 Identify and focus on the critical 20% that delivers most of the results.



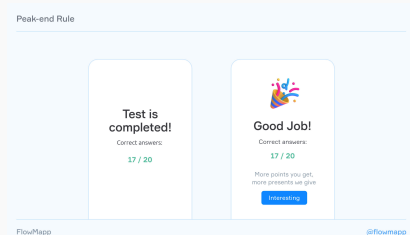
Parkinson's Law

- Any task will inflate until all of the available time is spent.
- Limit task completion time to user expectations.
- Reducing the actual time below the expected duration enhances user experience.
- Use features like autofill to expedite information entry.
- Examples include:
 - Quick form completions.
 - Efficient purchase and booking processes.
- Prevent task inflation and improve efficiency.



Peak-End Rule

- 1 Experiences are judged based on the peak and end moments.
- 2 Focus on creating memorable high points and a strong finish.
- 3 Example: A user experience with a memorable feature and positive final interaction.
- 4 Enhance key moments in user experiences to leave a lasting positive impression.



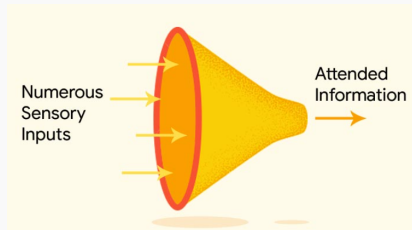
Postel's Law

- 1 Be liberal in what you accept, and conservative in what you send.
- 2 Systems should handle unexpected inputs gracefully.
- 3 Example: A form that accepts various input formats and provides clear error messages.
- 4 Design systems that accommodate a range of user inputs and handle errors effectively.



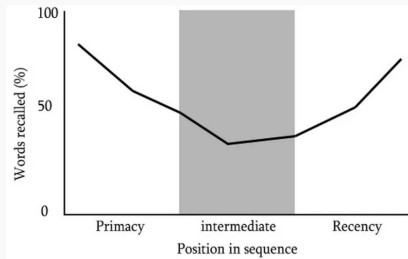
Selective Attention

- 1 Focusing attention on a subset of stimuli related to goals.
- 2 Users focus on what's relevant to their tasks or objectives.
- 3 Example: Highlighted notifications versus background information.
- 4 Design to emphasize and direct user focus on critical information.



Serial Position Effect

- 1 Users remember the first and last items in a series best.
- 2 The placement of information affects recall.
- 3 Example: Important features placed at the beginning and end of a list versus the middle.
- 4 Position key information at the beginning or end to enhance memory recall.



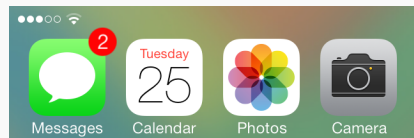
Tesler's Law

- 1 A certain amount of complexity is inherent in any system.
- 2 Some complexity cannot be eliminated and must be managed.
- 3 Example: A feature with unavoidable complexity, like advanced settings in a software.
- 4 Manage and present complexity in a user-friendly manner.



Von Restorff Effect

- 1 The item that differs from others is more likely to be remembered.
- 2 Distinctive items stand out and are easier to remember.
- 3 Example: A unique call-to-action button among similar ones.
- 4 Use distinctive elements to highlight important features or calls to action.



Zerigarnik Effect

- 1 People remember uncompleted or interrupted tasks better than completed tasks.
- 2 Unfinished tasks create a sense of urgency and stick in memory longer.
- 3 Example: A task manager that highlights incomplete tasks versus one that does not.
- 4 Use visual indicators to highlight and remind users of incomplete tasks.

